

PYTHON FUNDAMENTALS

INFO1 Mission 1 (S2)

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Installing Thonny

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<https://thonny.org/>

Binding INGINious

https://syllabus-interactif.info.ucl.ac.be/syllabus/info1-exercises/PART_I/MISSION_1/preparation

Missing from ingenious class

What is Python?



Programming languages are a way of telling the computer what to do.



High level is **simpler to write** (gets translated to lower level for executing)

Variables



- Direction is important (cannot be value-assignment-name)
- Variable names should **NOT** be keywords
- Variables are mutable and can **CHANGE** type
- Types are automatically **inferred**, **NOT DECLARED** (see next slide)

Values & Types

Example with inferred type.

<code>x = 5</code>	<code>#int</code>
<code>y = 3.14</code>	<code>#float</code>
<code>name = "Ada"</code>	<code>#string</code>
<code>is_ready = True</code>	<code>#boolean</code>

Expressions & Operators

PEMDAS... but use parentheses

```
X = 5 + 5 - 3 * 5
```

Use parentheses to remove confusion about order of operations.

Right-hand side evaluated first

```
X = 1  
X = x + 2
```

The right side is computed first, then stored.

! Operators can **CHANGE your type** without explicitly telling you:

```
X = 7 / 4 # x is now a float
```

Conditionals

- If / elif / else
- **!** Indentation matters: don't mix tabs and spaces (Thonny turns tabs into 4 spaces)
- A boolean statement decides which section runs
- **==** is **NOT** assignment, it's a conditional check
- Non-boolean values can act as true or false
- "and", "or", "not" combine boolean statements

```
if BOOLEAN_EXPRESSION:  
    STATEMENTS_1  # if True  
else:  
    STATEMENTS_2  # if False
```

```
X = 2  
A = 1  
X == (a + 1)  # True
```

```
x = 1  
print(x == True)  # True
```

While Loop

- Checked before **every pass**,
INCLUDING FIRST
- Make sure to update something in the loop to have an exit

```
count = 0
while count < 5:
    print(count)
    count += 1
```

Debugging: What Is It?

What does each type of error refer to?

1

Syntax errors

Code is written incorrectly.

2

Runtime errors

Errors that happen while the program runs.

3

Semantic errors

Code runs, but does the wrong thing.

RECAP

Variables, Values & Data Types

Variables & Types

- Different types
- Not explicit => type automatically inferred by Python interpreter

Assignment & Naming

- Assignment (right hand evaluated first)
- Careful with = and == and assignment direction
- Python variables are mutable
- Avoid keywords as variable names

Expressions & Operators

- Expression and operators
- PEMDAS -> use brackets

Conditionals & Loops

- Conditionals
- Boolean conditions
- Logical operators
- Conditional execution
- Conditional looping (while statement)

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Recap Done